

Understanding the Limitations of Machine Learning for IoT Anomaly Detection under Extreme Class Imbalance

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Abstract—The rapid growth of Internet of Things (IoT) devices has increased the need for effective network-based anomaly detection to identify malicious activities such as botnet communication and distributed attacks. While machine learning techniques have shown promising results in IoT intrusion detection, their effectiveness strongly depends on dataset quality and class distribution. This work investigates the impact of extreme class imbalance on IoT anomaly detection using the IoT-23 dataset. Network traffic features are extracted from packet capture data using fixed 30-second windows, and both statistical rule-based detection and Random Forest classification are evaluated. Experimental results show that rule-based methods tend to over-detect anomalies in highly variable malware traffic, while supervised machine learning models can achieve misleadingly high accuracy under severe class imbalance. Furthermore, class weighting and oversampling fail to adequately address the imbalance due to the limited diversity of minority-class samples. These findings demonstrate that dataset characteristics can significantly influence detection performance and highlight the need for more robust data collection, evaluation metrics, and learning strategies for reliable IoT forensic analysis.

Index Terms—IoT Security, Anomaly Detection, Machine Learning, Network Forensics, Class Imbalance

I. INTRODUCTION

The rapid proliferation of Internet of Things (IoT) devices has significantly transformed modern digital ecosystems, enabling smart homes, industrial automation, and connected healthcare systems. However, this growth has also introduced substantial security challenges due to the resource-constrained nature of IoT devices and their often limited built-in security mechanisms [1]–[3]. As a result, IoT networks have become attractive targets for large-scale cyberattacks, including botnet propagation, distributed denial-of-service (DDoS) attacks, and unauthorized data exfiltration [4].

Network-based anomaly detection has emerged as a promising approach for identifying malicious activities in IoT environments without requiring deep packet inspection. By analyzing traffic patterns such as packet counts, flow statistics, and temporal characteristics, anomaly detection systems can identify deviations from expected behavior [5]. These techniques are particularly valuable in encrypted communication scenarios, where payload inspection is not feasible [6].

Despite these advantages, anomaly detection in IoT environments faces several fundamental challenges. One of the primary limitations is the assumption of a stable baseline representing normal device behavior. In real-world IoT deployments, traffic patterns are typically consistent and predictable, enabling the construction of reliable statistical models. However, publicly available datasets used for research, such as IoT-23 [7], often contain predominantly malicious traffic with limited or no clean baseline data. This discrepancy raises concerns regarding the validity and generalizability of anomaly detection models trained on such datasets.

In this paper, we investigate the limitations of both rule-based and machine learning-based anomaly detection methods using the IoT-23 dataset. We extract statistical features from network traffic using fixed time windows and evaluate detection performance using a rule-based thresholding approach and a Random Forest classifier. Our results demonstrate that while rule-based methods tend to over-detect anomalies in highly variable traffic, supervised machine learning models fail to identify normal behavior due to insufficient representation of the minority class. Furthermore, we show that common techniques such as class weighting and oversampling do not fully address these challenges and may introduce overfitting. This work highlights the importance of data quality and class distribution in IoT datasets and demonstrates how these factors critically affect the reliability of machine learning-based anomaly detection systems.

The main contribution of this work is the analysis of how extreme class imbalance and dataset characteristics can lead to misleading anomaly detection performance in IoT-23.

II. RELATED WORK

IoT security has become an active area of research due to the increasing number of connected devices and their vulnerability to cyberattacks. Several studies have explored network-based approaches for detecting anomalous behavior in IoT environments. Early work by Ammar *et al.* [1] provides a comprehensive survey of IoT security challenges, highlighting the lack of robust protection mechanisms in resource-constrained devices.

One prominent approach to IoT anomaly detection is device fingerprinting based on network traffic patterns. Meidan *et al.* [5] proposed *ProfilIoT*, which uses statistical features extracted from network flows to identify IoT devices and detect anomalous behavior without inspecting packet payloads. Similarly, Acar *et al.* [6] demonstrated that encrypted IoT traffic still leaks behavioral patterns, enabling inference of user activities through traffic analysis.

The availability of realistic datasets has played a crucial role in advancing IoT intrusion detection research. The IoT-23 dataset introduced by Garcia *et al.* [7] is one of the most widely used benchmarks, containing labeled network traffic from various IoT malware scenarios, including botnet activity and command-and-control communication. In addition, other datasets such as TON_IoT [8] have been developed to support the evaluation of AI-based security systems in IoT environments, providing diverse scenarios for intrusion detection research. Studies using IoT-23 have applied both classical machine learning and deep learning models to detect malicious traffic. Several studies have explored machine learning techniques for IoT intrusion detection, including DDoS detection and traffic classification [3], [9], [10]. For example, several works have reported high classification accuracy using algorithms such as Random Forests, Support Vector Machines, and neural networks. However, these results often rely on datasets with highly imbalanced class distributions, which may lead to misleading performance metrics.

Deep learning approaches, including autoencoders and neural networks, have also been applied to intrusion detection tasks [11], [12]. However, these methods require large labeled datasets and significant computational resources, which may limit their applicability in real-world IoT environments.

Unsupervised and semi-supervised approaches, such as Isolation Forest and density-based methods, have also been explored for anomaly detection [13], [14]. However, their effectiveness depends on the availability of a reliable baseline, which is often lacking in public IoT datasets.

A critical but less explored issue in existing literature is the impact of dataset characteristics on anomaly detection performance. In particular, extreme class imbalance and the absence of representative normal traffic can significantly degrade model performance. Although methods such as class weighting and oversampling are commonly used to mitigate imbalance, they may introduce overfitting and fail to capture meaningful patterns when the minority class lacks diversity. Class imbalance is a well-known challenge in machine learning, significantly affecting model performance and evaluation reliability [15], [16].

In contrast to prior work that primarily focuses on improving detection accuracy, this study emphasizes the limitations of both rule-based and machine learning approaches when applied to highly imbalanced IoT datasets. While numerous studies have reported high detection accuracy using IoT-23, most focus on classifier performance rather than dataset characteristics. By analyzing the IoT-23 dataset, we demonstrate that extreme class imbalance and the absence of representative

benign traffic can produce misleadingly high performance while limiting generalization to realistic deployment scenarios. These findings highlight the importance of robust evaluation metrics, balanced datasets, and data-aware detection strategies in IoT anomaly detection research.

III. METHODOLOGY

This section describes the dataset, feature extraction process, anomaly detection methods, and evaluation strategy used in this study.

A. Dataset

The experiments in this work are conducted using the IoT-23 dataset [7], which contains labeled network traffic captures of IoT devices under various malware scenarios. Specifically, we use the *CTU-IoT-Malware-Capture-1-1* scenario, which represents botnet-infected IoT traffic exhibiting behaviors such as scanning, command-and-control (C2) communication, and burst transmissions.

Traffic was aggregated into fixed 30-second windows for analysis. Window labels were assigned based on the packet labels contained within each aggregation interval, with a window labeled malicious if it contained malicious traffic and benign otherwise. After aggregation, the dataset contained 13,419 malicious windows and 10 benign windows, corresponding to more than 99.9% malicious traffic. This extreme class imbalance presents a significant challenge for both anomaly detection and machine learning-based classification methods.

B. Feature Extraction

To analyze network behavior, raw PCAP files are converted into structured data using *TShark*. Traffic is aggregated into fixed 30-second time windows to capture temporal characteristics while maintaining computational efficiency.

For each window, statistical features including packet count, bytes transmitted, number of unique destination IPs and ports, and inter-arrival time variance are extracted.

These features are selected to capture both volumetric and temporal aspects of network traffic, which are indicative of anomalous behavior in IoT environments.

C. Rule-Based Anomaly Detection

A statistical rule-based anomaly detector is used as a baseline. For each feature, the mean and standard deviation are computed from an assumed baseline period. A window is flagged as anomalous if $X > \mu + k\sigma$, where μ and σ are the feature mean and standard deviation, respectively, and k is a sensitivity parameter. This method assumes a stable baseline, which may not exist in malware traffic.

D. Machine Learning-Based Detection

To improve detection performance, a supervised machine learning model based on Random Forest is employed. The classifier is trained using the extracted feature set described above.

Due to the extreme class imbalance in the dataset, a stratified train-test split is first performed to preserve the class

distribution. Subsequently, oversampling is applied only to the training data to balance the minority class. This is achieved by randomly replicating instances of the minority class until both classes have equal representation.

The Random Forest model is trained with multiple decision trees, allowing it to capture non-linear relationships between features. Feature importance is also extracted to identify the most significant indicators of anomalous behavior.

In addition, confusion matrices are used to analyze classification performance in detail. Given the highly imbalanced nature of the dataset, particular attention is paid to recall and F1-score, as accuracy alone can be misleading.

E. Evaluation Metrics

The performance of both methods is evaluated using accuracy, precision, recall, F1-score, and confusion matrices. Given the highly imbalanced dataset, recall and F1-score are emphasized because accuracy alone can be misleading.

F. Experimental Setup

All experiments were conducted in Python using Pandas, Scikit-learn, and Matplotlib. The Random Forest classifier used 100 decision trees with a fixed random seed. Oversampling was applied only to the training data to prevent data leakage and preserve realistic evaluation conditions.

IV. RESULTS AND DISCUSSION

A. Traffic Behavior Analysis

Fig. 1 shows the packet count per 30-second window for the CTU-IoT-Malware-Capture-1-1 dataset. The traffic exhibits highly bursty behavior, with several sharp spikes exceeding 1600 packets. These bursts are indicative of malicious activities such as botnet scanning, distributed denial-of-service (DDoS) attempts, and command-and-control (C2) communications. Similar trends are observed in transmitted bytes, confirming that malware traffic exhibits both bursty behavior and high data volume.

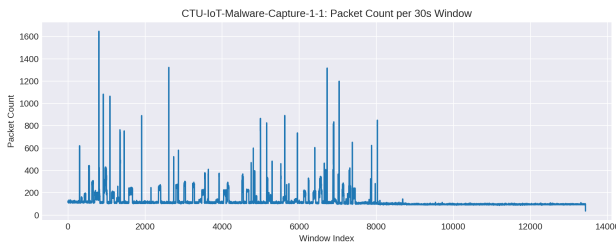


Fig. 1. Packet count per 30-s window.

These observations indicate that IoT-23 malware traffic lacks a stable baseline, unlike typical benign IoT environments.

B. Rule-Based Anomaly Detection

Fig. 2 illustrates the anomalous windows detected using a rule-based statistical thresholding approach. The red markers indicate windows flagged as anomalies based on deviations from the mean and standard deviation of baseline features.

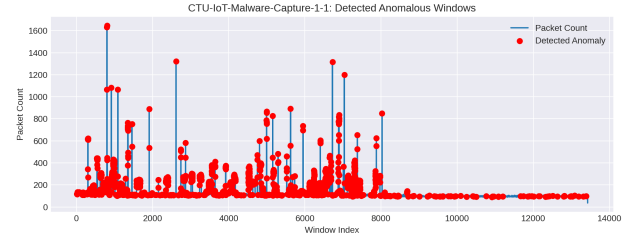


Fig. 2. Detected anomalous windows using rule-based statistical thresholding.

A large proportion of windows are classified as anomalous, reflecting the highly irregular nature of the malware traffic. This behavior indicates that simple statistical methods are overly sensitive in such environments and are unable to distinguish between normal variability and true malicious activity. As a result, the rule-based approach demonstrates high precision but low recall, missing subtle anomalies while over-detecting high-variance regions.

C. Machine Learning Performance

A Random Forest classifier was trained using extracted features, including packet count, bytes transmitted, number of unique destination IPs, number of destination ports, and inter-arrival time variance. Feature importance is shown in Fig. 3.

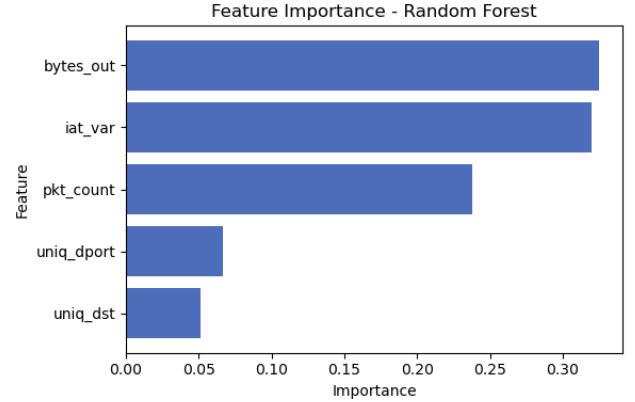


Fig. 3. Feature importance derived from the Random Forest model.

The results indicate that *bytes_out* and *iat_var* are the most influential features, highlighting the importance of traffic volume and temporal variability in detecting anomalous behavior.

The confusion matrix obtained after applying oversampling is shown in Fig. 4. Table I highlights the misleading nature of accuracy under extreme class imbalance.

Although the confusion matrix appears to show perfect classification performance after oversampling, this result should be interpreted with caution. The minority normal class contains very few unique samples, and oversampling replicates these instances, leading to overfitting. Therefore, the apparent performance is likely inflated and does not reflect true generalization capability.

D. Discussion

The experimental results highlight several critical challenges in IoT anomaly detection. First, the absence of a stable

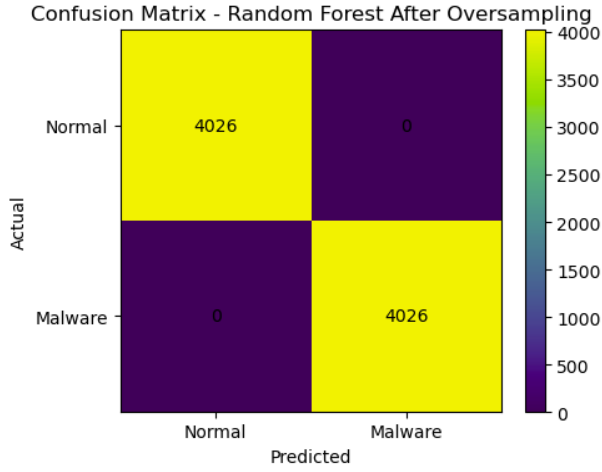


Fig. 4. Confusion matrix of the Random Forest model after oversampling.

TABLE I
PERFORMANCE COMPARISON OF DETECTION METHODS

Method	Accuracy	Precision	Recall	F1-score
Rule-based	0.16	1.00	0.16	0.28
Random Forest	0.99	0.99	1.00	0.99

baseline in malware datasets such as IoT-23 limits the effectiveness of statistical anomaly detection methods. Second, extreme class imbalance severely impacts supervised learning models, causing them to favor the majority class. Third, naive application of oversampling techniques can lead to artificially inflated performance due to overfitting.

These findings demonstrate that accuracy alone is not a reliable performance metric in highly imbalanced IoT datasets. Instead, careful consideration of data distribution, feature diversity, and evaluation metrics is essential. The results emphasize the need for more robust approaches, such as unsupervised or semi-supervised learning, for real-world IoT forensic analysis. The present study focuses on a single IoT-23 malware capture and therefore does not claim generalization across all IoT malware scenarios. Nevertheless, the findings demonstrate how extreme class imbalance can distort evaluation results and motivate broader validation across additional IoT datasets. Practical deployment of IoT anomaly detection systems should emphasize the collection of representative benign traffic, protocol-aware feature engineering, and semi-supervised or unsupervised learning approaches when labeled baseline data are limited.

V. CONCLUSION AND FUTURE WORK

In this work, we evaluated rule-based and machine learning-based anomaly detection methods for IoT traffic using the IoT-23 dataset. Results show that rule-based methods over-detect anomalies due to high traffic variability, while Random Forest achieves near-perfect accuracy that is misleading under extreme class imbalance. Oversampling techniques were found to introduce overfitting due to the lack of diversity in minority class samples. These findings demonstrate that dataset characteristics, rather than model choice, are the primary limitation in IoT anomaly detection. These findings highlight

the importance of dataset quality and robust evaluation metrics for IoT forensic systems.

Future work will focus on collecting more balanced IoT datasets, incorporating protocol-aware traffic features, and exploring unsupervised or semi-supervised anomaly detection methods that are less dependent on labeled data. In addition, the analysis will be extended to multiple IoT-23 captures and other public IoT security datasets to evaluate the generality of the observed class imbalance effects and to develop more robust evaluation strategies beyond accuracy.

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REFERENCES

- [1] M. Ammar, G. Russello, and B. Crispo, "Internet of Things: A Survey on the Security of IoT Frameworks," *Journal of Information Security and Applications*, vol. 38, pp. 8–27, 2018.
- [2] S. Sicari, A. Rizzardi, L. A. Grieco, and A. Coen-Porisini, "Security, privacy and trust in Internet of Things: The road ahead," *Computer Networks*, vol. 76, pp. 146–164, 2015.
- [3] M. Conti, A. Dehghantanha, K. Franke, and S. Watson, "Internet of Things security and forensics: Challenges and opportunities," *Future Generation Computer Systems*, vol. 78, pp. 544–546, 2018.
- [4] C. Koliadis, G. Kambourakis, A. Stavrou, and J. Voas, "DDoS in the IoT: Mirai and Other Botnets," *Computer*, vol. 50, no. 7, pp. 80–84, 2017.
- [5] Y. Meidan, M. Bohadana, A. Shabtai, et al., "ProfilIoT: Profiling IoT Devices to Detect Anomalous Behavior," *arXiv preprint arXiv:1708.09890*, 2017.
- [6] G. Acar et al., "Peek-a-Boo: I See Your Smart Home Activities, Even Encrypted!," in *Proc. ACM WiSec*, 2020, pp. 207–218.
- [7] S. Garcia, A. Parmisano, and M. J. Erquiaga, "IoT-23: A Labeled Dataset with Malicious and Benign IoT Network Traffic," *Zenodo*, 2020, doi:10.5281/zenodo.4743746.
- [8] N. Moustafa, "A new distributed architecture for evaluating AI-based security systems at the edge: Network TON_IoT datasets," *Future Generation Computer Systems*, 2021.
- [9] R. Doshi, N. Aphorpe, and N. Feamster, "Machine learning DDoS detection for consumer Internet of Things devices," in *Proc. IEEE S&P Workshops*, 2018.
- [10] T. T. Nguyen and G. Armitage, "A survey of techniques for Internet traffic classification using machine learning," *IEEE Communications Surveys & Tutorials*, 2008.
- [11] K. Alrawashdeh and C. Purdy, "Toward an online anomaly intrusion detection system based on deep learning," in *Proc. IEEE ICMLA*, 2016.
- [12] N. Shone, T. N. Ngoc, V. D. Phai, and Q. Shi, "A deep learning approach to network intrusion detection," *IEEE Transactions on Emerging Topics in Computational Intelligence*, 2018.
- [13] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation Forest," in *Proc. IEEE ICDM*, 2008.
- [14] M. M. Breunig et al., "LOF: Identifying density-based local outliers," in *Proc. ACM SIGMOD*, 2000.
- [15] H. He and E. A. Garcia, "Learning from imbalanced data," *IEEE Transactions on Knowledge and Data Engineering*, vol. 21, no. 9, pp. 1263–1284, 2009.
- [16] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic minority over-sampling technique," *Journal of Artificial Intelligence Research*, 2002.